

WiSe 2026 : Research Presentation

Name: Marie Ange Malla Ndassi

Affiliation: African Institute for Mathematical Sciences (AIMS) Cameroon

Presentation Title:

Assessing the impact of training data composition on deep learning-based cardiac structure segmentation in echocardiography

Abstract

The accurate segmentation of cardiac anatomical structures in echocardiography is essential for reliable cardiac function evaluation and clinical decision-making, yet the performance of deep learning models for this purpose is highly sensitive to the size and composition of the training data. This thesis investigates the impact of different training data compositions on deep learning-based framework for cardiac structure segmentation in echocardiography, focusing on the left ventricle, myocardium, and left atrium. A CycleGAN-based style transfer pipeline is built to synthesize labelled echocardiographic image-mask pairs by mapping the visual appearance of EchoNet-Dynamic images onto CAMUS frames, producing what we refer to as pseudo-CAMUS data. Three segmentation architectures : U-Net, PMFSNet, and Swin UNETR are trained and evaluated under four experimental conditions: real data only, real data with classical augmentation, GAN-generated data only, and a combination of real and GAN-generated data. An additional evaluation of ejection fraction estimation from the predicted masks is performed. The results demonstrate that training data composition significantly impacts segmentation performance in an architecture-dependent manner. U-Net achieves the strongest overall results, reaching a left ventricle (LV) Dice of 0.9322 under the baseline condition, and proves the most robust across all configurations. Swin UNETR benefits most from classical augmentation, improving its LV Dice from 0.8762 to 0.8995 under this condition. PMFSNet is the most sensitive to data composition, experiencing near-complete failure under GAN-generated data alone, with a left atrium Dice of only 0.0613 and a Hausdorff distance of 139.67 mm but recovering to its best performance under the combined strategy, reaching a LV Dice of 0.8945. GAN-generated data alone consistently degrades performance across all models, while its combination with real annotations yields tangible improvements in most configurations. Regarding ejection fraction estimation, U-Net under the combined condition achieves the best results with a mean absolute error of 5.79 that segmentation accuracy and ejection fraction estimation reliability do not always correlate, indicating that Dice-based evaluation alone may be insufficient for assessing clinically meaningful performance.

Keywords: Cardiac structure segmentation, echocardiography, cycleGAN, data augmentation, deep learning.

Biography

Marie Ange Malla Ndassi is a Data Science student at the African Institute for Mathematical Sciences (AIMS) Cameroon, with a background in physics and electrical engineering. Her research interests have increasingly focused on machine learning and computer vision, particularly their applications to medical imaging. In 2026, she participated in the G-RIPS research program at Tohoku University in Japan, where she worked on computational imaging and image reconstruction, further developing her experience in computer vision and computational

methods. Her recent research focuses on generative models and semi-supervised learning for cardiac image segmentation, with an emphasis on developing robust methods for settings where annotated data are limited.